WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

# Simulation for Data Science

Bootstrap, Monte Carlo, permutation test



# The big idea: let the computer repeat it

| | |
|---|---|
| **1 Mimic** | Define a random process that imitates the situation |
| **2 Repeat** | Run it many times: 1,000 to 100,000 is typical |
| **3 Read** | Look at the pile of results as a histogram |

The spread of the results is the answer. No formula needed.

# Three questions, three methods

| Method | Question it answers | Where the randomness comes from |
|---|---|---|
| **Bootstrap** | How uncertain is my estimate? | Resampling my data with replacement |
| **Monte Carlo** | What could happen, and how likely? | Drawing from a probability model |
| **Permutation test** | Is this difference more than luck? | Shuffling the group labels |

Same recipe each time: repeat, collect, read the histogram.

# Bootstrap: resample your data

| | |
|---|---|
| **1 Start** | One sample of n rows |
| **2 Resample** | Draw n rows with replacement |
| **3 Compute** | Average of the resample |
| **4 Repeat** | 1,000 times |
| **5 Read** | The spread of the 1,000 averages |

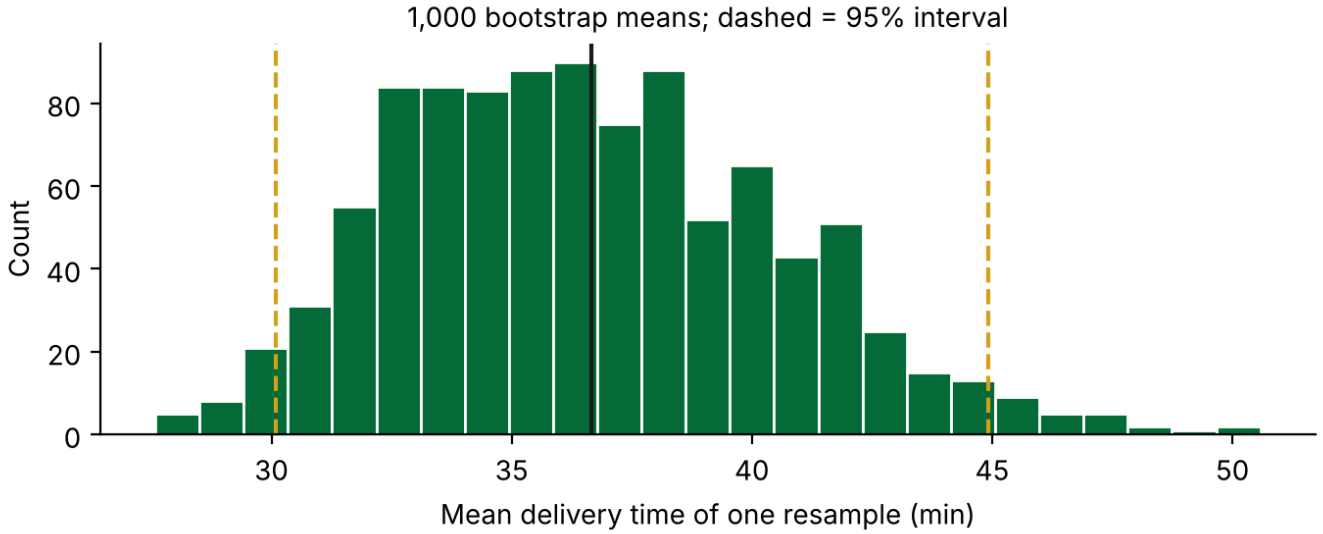
Idea: the spread of resampled averages shows how much an average can vary.

# Resampling with replacement

| Resample | Values drawn | Average |
|---|---|---|
| **Original sample** | 22, 25, 30, 44, 75 | 39.2 |
| **Resample 1** | 75, 22, 22, 25, 22 | 33.2 |
| **Resample 2** | 75, 75, 30, 22, 22 | 44.8 |
| **Resample 3** | 25, 30, 44, 30, 25 | 30.8 |

Same size, values repeat, some are left out: so every average differs.

# Example: 12 delivery times




Middle 95% of resampled averages: 30.1 to 44.9 minutes.

# Reading the bootstrap result

**Confidence interval**

The middle 95% of the resampled averages: 30.1 to 44.9 minutes.

**Lopsided on purpose**

6.6 below the mean, 8.3 above. The long tail of slow deliveries stretches the upper side.

**Central Limit Theorem**

Averages form a bell shape even when the raw data are skewed. The histogram shows it.

The bootstrap lets the data decide the shape of the interval.

# When the bootstrap struggles

| Problem | Why it breaks |
| --- | --- |
| **Biased sample** | Resampling copies the bias. Only weekday data says nothing about weekends. |
| **Very small sample** | Few distinct values, so resamples look alike and the interval is unstable. |
| **Extreme values** | A resample can never exceed the largest value in the sample (e.g. the maximum). |

Simulation cannot repair a bad sample.

# Monte Carlo: play the model many times

| | |
|---|---|
| **1 Model** | Write down how the world works, with its probabilities |
| **2 Simulate** | Draw one possible outcome from the model |
| **3 Repeat** | Draw thousands of outcomes |
| **4 Count** | How often did the event of interest happen? |

Probability = share of simulated runs in which the event happened.

# Example: failure of a bracket under load

**Load**

Force on the bracket. Normal: mean 11 kN, standard deviation 1.8 kN.

**Structure**

Plate thickness. Normal: mean 5.0 mm, standard deviation 0.15 mm (tolerance).

**Material**

Yield strength. Median 250 MPa, about 6% batch-to-batch scatter.

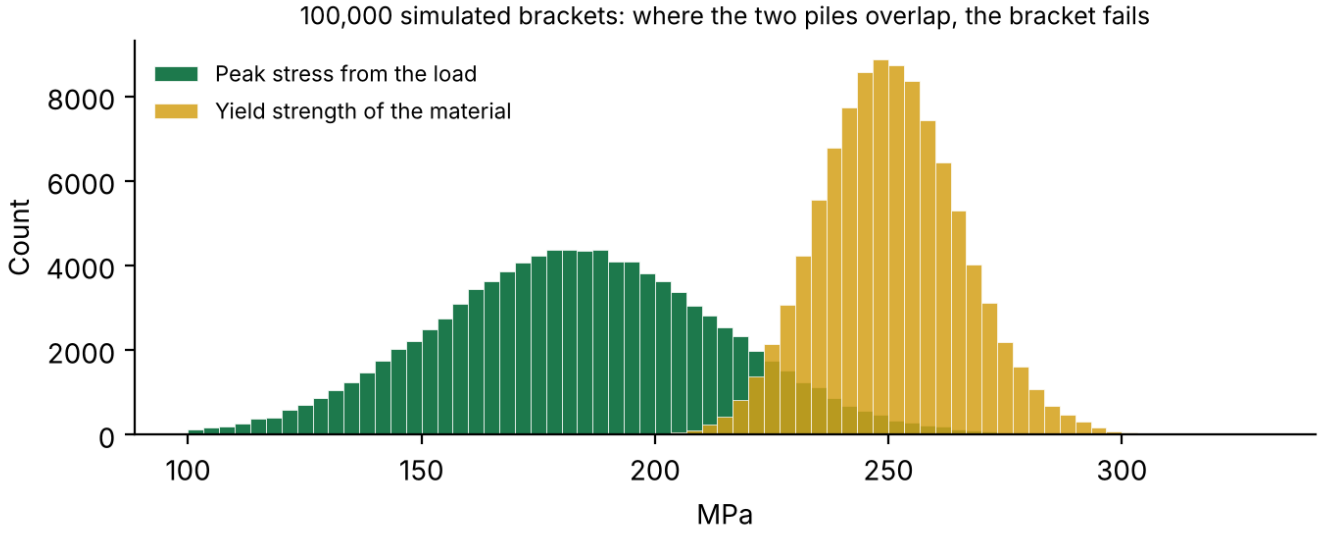
FEA gives the peak stress. Failure = peak stress above yield strength.

# One FEA run = one simulated bracket

| Bracket | Load (kN) | Thickness (mm) | Yield (MPa) | Peak stress (MPa) | Result |
|---|---|---|---|---|---|
| **A** | 11.1 | 5.11 | 238 | 180 | Safe |
| **B** | 13.4 | 4.95 | 268 | 227 | Safe |
| **C** | 12.7 | 4.88 | 212 | 217 | Fails |
| **D** | 10.1 | 4.91 | 245 | 171 | Safe |

Bracket C fails because three moderate effects line up: high load, thin plate, weak batch.

# Result: 100,000 simulated brackets




2,468 of 100,000 brackets failed: a failure probability of about 2.5%.

# Which input drives the failure?

| Input | Type | Share of variation in safety margin (approx.) |
|---|---|---|
| **Load** | Loading | about 78% |
| **Yield strength** | Mechanical property | about 19% |
| **Thickness** | Structure | about 3% |

Simulation also tells you where to spend effort: here, on the load.

# The cost problem: surrogate models

| Approach | FEA runs | Failure probability |
|---|---|---|
| **Direct Monte Carlo** | 100,000 | 2.5% |
| **Surrogate: linear regression** | 200 | 2.4% |
| **Surrogate: random forest** | 200 | 2.0% |

A few hundred FEA runs plus a surrogate can replace 100,000 solver runs.

# Monte Carlo trusts its model

| Assumption in the model | What if it is wrong? |
|---|---|
| **Input distributions are right** | A wrong load or strength distribution gives a wrong failure probability. |
| **Inputs are independent** | If they are linked, extreme combinations are more likely than simulated. |
| **The FEA model is right** | A poor mesh, material law or boundary condition affects every run. |

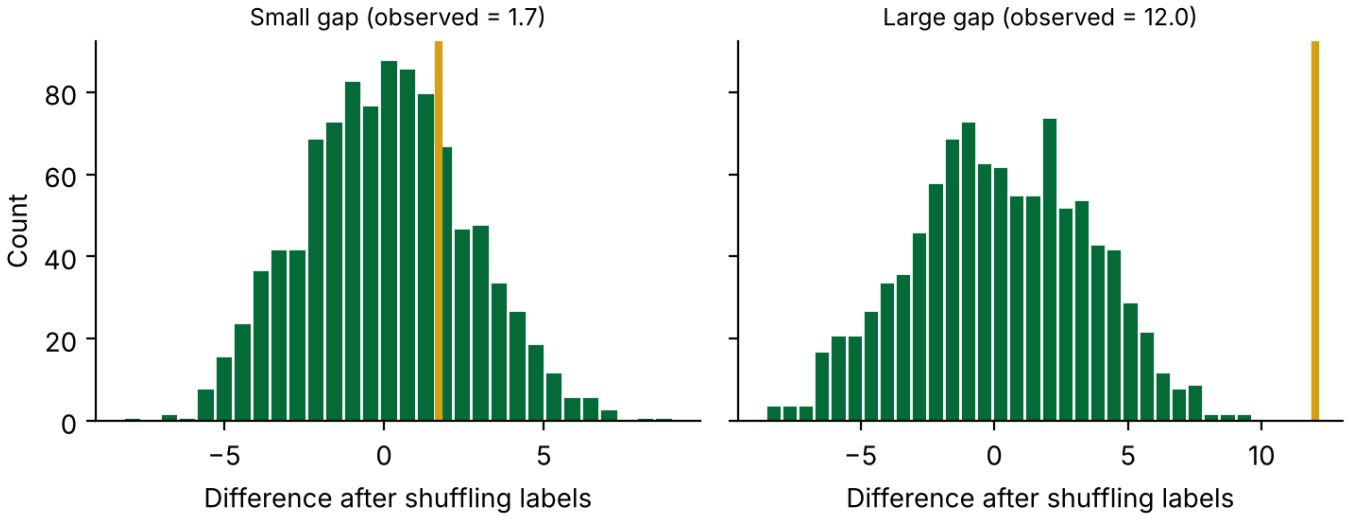
Change the assumptions, run again, and see how the answer moves.

# Permutation test: is it just luck?

| | |
|---|---|
| **1 Claim** | The label has no effect on the number |
| **2 Shuffle** | Randomly reassign the labels to the rows |
| **3 Recompute** | The difference in group averages |
| **4 Repeat** | 1,000 times: this is what luck produces |
| **5 Compare** | Where does the real difference fall? |

If the real gap rarely appears in the shuffled pile, luck is a poor explanation.

# Example: salaries in two groups




Small gap: seen in 51% of shuffles (luck). Large gap: in 0 of 1,000 (real).

# What a p-value does and does not say

**p-value**

Share of shuffles with a gap as big as the real one. Small means unlikely to be luck.

**Not said**

Not how big the gap is, not whether it matters, not that the label causes it.

**Link to features**

Shuffle a feature column. If the model gets worse, the model relied on it.

Statistically clear is not the same as practically important.

# Comparing the three methods

| | Bootstrap | Monte Carlo | Permutation test |
|---|---|---|---|
| **Question** | How uncertain is my estimate? | What outcomes are possible and how likely? | Is this difference more than luck? |
| **Randomness from** | Resampling the data | Drawing from a probability model | Shuffling the group labels |
| **Result** | Confidence interval | A probability or a range of outcomes | A p-value |

Pick the method by the question you are asking.

# Pros: Monte Carlo, regression, other ML

| Method | Pros |
|---|---|
| **Monte Carlo with FEA** | Real physics: works for new designs with no test data. Any input distribution. Gives a failure probability and what drives it. |
| **Linear regression** | Needs few runs. Fast. Coefficients are easy to read. Works well when the response is close to a straight line. |
| **Other ML (random forest, neural network, Gaussian process)** | Captures curves and interactions. Very fast once trained. Handles many inputs. |

Physics-based Monte Carlo is trusted for new designs; regression and ML are fast.

# Cons: Monte Carlo, regression, other ML

| Method | Cons |
|---|---|
| **Monte Carlo with FEA** | Each sample is a solver run (minutes to hours). Rare failures need huge samples. Only as good as the model and inputs. |
| **Linear regression** | Misses curves, interactions and sudden failure. Extrapolates badly. Tail errors are hidden. |
| **Other ML (random forest, neural network, Gaussian process)** | Needs good training runs. Can be a black box. Unreliable outside the training range. Hard to check in the tail. |

In practice: train a surrogate on FEA runs, then run Monte Carlo on the surrogate.

# Key points

| | |
|---|---|
| **Simulation** | Replace a formula with repetition, then read the histogram |
| **Bootstrap** | Resample the data to see how much an estimate can vary |
| **Monte Carlo** | Play a model of the world many times to estimate a probability |
| **Permutation** | Shuffle labels to see what luck looks like, then compare |
| **Caution** | A biased sample or a wrong model gives a wrong answer |

More runs make the answer smoother, not more correct.